



The Anita's Veg Café Data Saga

*From a Home Kitchen in Bengaluru to a Full-Blown Business Intelligence
Investigation*

*"Every empire starts with a single plate of Paneer Butter
Masala."*

— probably not Anita, but she'd agree

P Prologue — The Spark

Bengaluru, early 2021. The city is loud with delivery bikes and louder with hunger.

Somewhere in a modest kitchen, **Anita** — a home cook with more passion than capital — decides she's done cooking just for family. She opens the doors of a tiny vegetarian café and, without knowing it, plants the seed of what will eventually become a **data-driven business empire**.

She has no dashboards. No spreadsheets. No SQL. Just three recipes, a handful of plastic chairs, and an instinct that homely food, done right, sells itself.

She was right. But instinct can only take a business so far — the rest, as you're about to see, takes **data**.

1 The Café That Refused to Stay Small

Word spreads fast in a city full of hungry young professionals and broke-but-loyal college students. Anita's café becomes the unofficial lunch headquarters for two very different tribes:

The Hustlers

Young professionals craving a quick, honest meal between meetings.

The Scholars

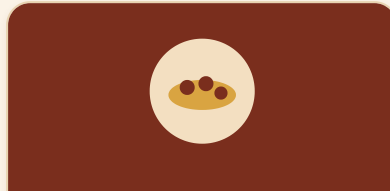
College students hunting for affordable, filling comfort food.

What unites them? **Homely food. Comfort in every bite.** No fusion gimmicks, no influencer plating — just food that tastes like someone's mother made it, because someone's mother basically did.

2

The Holy Trinity of the Menu

Anita doesn't need fifty items on her menu. She has **three**, and each one has earned its title through pure battlefield performance:



👑 BEST SELLER

Paneer Butter Masala

₹180

Rich, creamy, and permanently in demand — the dish regulars order without even glancing at the menu.



🍷 SHAREABLE LEGEND

Veg Biryani

₹150

Aromatic, generous, built for splitting between friends who "just want a small plate" and then don't stop eating.



🥳 ALL-DAY WARRIOR

Masala Dosa

₹120

A South Indian classic that never clocks out — it sells from breakfast to dinner without ever losing momentum.

Three dishes. Zero filler. This is a menu built on conviction, not variety.

3

The Loyalty Uprising

Success breeds regulars, and regulars start talking. A handful of them — the ones who show up so often the staff know their order before they sit down — plant an idea in Anita's head:

"Why not reward us for coming back?"

And so, **the loyalty program is born**. Every rupee spent starts converting into points. Customers who join aren't just eating anymore — they're **investing** in the café, one Masala Dosa at a time.

But a loyalty program isn't just a reward system. It's a **data-generation machine**. Every join date, every post-loyalty order, every point earned becomes a thread Anita can pull to understand her business at a level gut instinct could never reach.

4

The Data Awakening

Here's where the story shifts from *restaurant origin tale* to *business intelligence thriller*. Anita starts asking questions her instincts can't answer:

🔥 Which dishes are actually driving the business — not just the ones that "feel" popular?

↻ Is the loyalty program actually changing how people behave, or is it just a shiny badge?

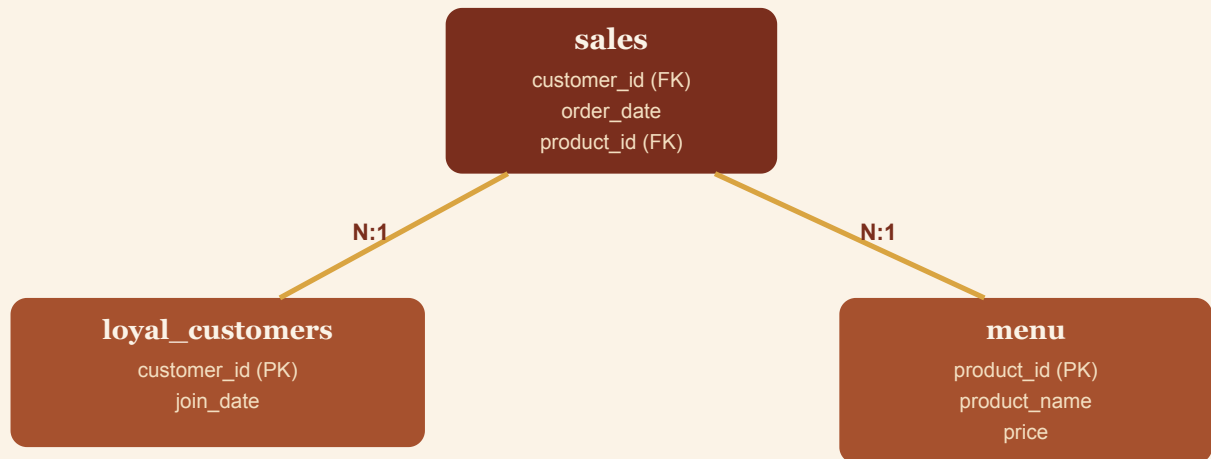
🚀 What's the next move — and how do I know it's the right one, not just a guess?

She doesn't have an analytics team. She doesn't have a BI platform. What she has is **raw transactional data** — and she needs someone who can turn rows of numbers into decisions. That's where **you** — her data consultant — step in.

5

The Architecture Behind the Aroma

Before any investigation begins, every detective needs a map. Here's the blueprint of Anita's business, distilled into three interconnected tables living under the schema `anitas_veg_cafe`:



📅 sales

The diary of every bite ever taken. Every row is a moment: someone, on some day, ordered something. This is the pulse of the café.

🍽️ menu

The identity card of every dish. Turns a faceless `product_id` into a name and a price.

👤 loyal_customers

The roll call of the converted — their `join_date` is the dividing line between life before and after loyalty.

Sample evidence from the field:

customer_id	order_date	product_id
Aarav	2021-01-01	1
Aarav	2021-01-01	2
Aarav	2021-01-07	2
Meera	2021-01-01	2
Meera	2021-02-01	3
Rohan	2021-01-01	3

product_id	product_name	price
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customer_id	join_date
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1	Paneer Butter Masala	180	Aarav	2021-01-07
2	Veg Biryani	150	Meera	2021-01-09
3	Masala Dosa	120		

6 The Ten Trials

Anita doesn't just want numbers. She wants **a story her data can tell her** — one mission at a time. The ten questions she hands you aren't a checklist. They're **three investigations**, each one peeling back a different layer of her business.



Mission 01 — Spending & Visits

The Money Trail

Trial 1 — The Wallet Audit

What is the total amount each customer has spent at the café?

Every rupee, tracked back to the person who spent it — Anita's first real look at her highest-value customers.

Trial 2 — The Footprint Count

How many distinct days has each customer placed an order?

Not orders — days. A customer who visits 10 different days is a different kind of loyal than one who orders 10 items in one sitting.



Mission 02 — Product Preferences

The Taste Test

Trial 3 — The First Impression

What was the first dish ordered by each customer?

The gateway order — the dish that hooked them.

Trial 4 — The Crowd Favorite

Which menu item is the most popular overall?

The undisputed champion carrying the café on its back.

Trial 5 — The Personal Obsession

What is the most frequently ordered dish for each customer?

Popularity isn't universal — every regular has their dish.

Trial 6 — The First Loyal Bite

After joining the loyalty program, what dish did each member first order?

The moment loyalty turned from a decision into a real, edible action.

Trial 7 — The Farewell Order

Before joining the loyalty program, what dish did each customer order last?

A snapshot of the moment right before commitment.

Trial 8 — The Pre-Loyalty Ledger

For each member, how many items and how much did they spend before joining?

A baseline — you can't measure loyalty's impact without knowing what came before it.

Trial 9 — The Points Formula

If ₹1 = 10 points, and Paneer Butter Masala earns double points, how many points does each customer earn?

Currency into loyalty currency, with a built-in incentive twist.

Trial 10 — The Double-Points Window

In their first loyalty week, members earn double points on all items. How many points do Aarav and Meera have by the end of January?

The most intricate trial — a time-boxed bonus window layered on top of everything else.

To crack these ten trials, this isn't a job for basic `SELECT` *. This investigation demands the full SQL arsenal:

Technique	Where It's Used
JOINS	Connecting sales ↔ menu ↔ loyal_customers to turn IDs into meaning
Aggregations (SUM, COUNT, AVG)	Trials 1, 2, 4, 8 — totals, counts, and spend
Window Functions (ROW_NUMBER, RANK)	Trials 3, 5, 6, 7 — finding "first," "last," and "most frequent" per customer
CASE Statements	Trial 9 — the double-points logic for Paneer Butter Masala
Date Logic (DATEDIFF, date ranges)	Trial 10 — the first-loyalty-week bonus window

E Epilogue — What This Project Really Proves

This isn't just a café analyzing sales. It's a **complete, self-contained Business Intelligence case study**, compressed into three tables and ten questions:

✓ **Relational data modeling** — understanding how a normalized schema connects real business entities

✓ **End-to-end SQL fluency** — joins, aggregation, window functions, conditional logic, and date arithmetic, all in one project

✓ **Business translation** — converting a business owner's plain-language curiosity into precise analytical queries

✓ **Loyalty & behavioral analytics** — a genuinely advanced pattern (before/after impact, time-boxed bonus logic) mirroring real-world CRM and retention analysis

What started as one woman's home-cooked dream became a fully-fledged analytics investigation — proof that you don't need 300,000 rows to do *real* data science. Sometimes, all it takes is three tables, ten sharp questions, and someone willing to actually ask them.

 **Anita's Veg Café** 

Where homely flavors meet data-driven insight.